

# Subramanian Veerappan

Process Design Engineer

Randers, Denmark

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## Professional Summary

Process Design Engineer with hands on experience in process simulation, P&ID review, and FEED deliverables across greenfield and operational pyrolysis plant projects. Supported PSV revalidation to API 520/521, nitrogen blanketing calculations, and Python based root cause analysis on large scale SCADA operational data. My MSc in Thermal and Process Engineering grounds the thermodynamic side; my independent portfolio work covers equipment sizing, flare system sizing using Aspen Flare Analyzer, and full P&ID development built on my own initiative. I pick things up fast, I ask the right questions, and I do not let gaps stay gaps.

*Portfolio and engineering deliverables:* [subbu1310.github.io](https://subbu1310.github.io)

## Core Skills

- **Thermodynamics and Thermal Processes:** MSc in Thermal and Process Engineering; pyrolysis plant process simulation and heat and mass balance analysis
- **Process Design:** BFD to PFD progression, P&ID review and coordination, equipment datasheets, FEED deliverables, stream tables; full P&ID development (independent portfolio)
- **Process Safety:** PHA facilitation and participation (HAZOP, FMEA, FTA, What-If, Checklist); OSHA PSM Rule; PSV and flare system sizing (API 520/521/527, Aspen Flare Analyzer)
- **Equipment Sizing:** Pump sizing, control valve sizing (ISA 75.01 / IEC 60534), heat exchanger thermal sizing (TEMA), hydraulic calculations and pressure drop analysis
- **Data Analysis and Root Cause Analysis:** Python-based SCADA time series analysis, downtime pattern identification, automated reporting workflows, Excel, MATLAB
- **Simulation:** DWSIM, Aspen HYSYS, Aspen Flare Analyzer, FluidFlow
- **Standards and Compliance:** API 520/521/527, TEMA, ISA 75.01 / IEC 60534, OSHA 29 CFR 1910.119
- **CAD and Drafting:** AutoCAD Plant 3D (including 3D piping layout and model review), Microsoft Visio; isometric checking

## Professional Experience

### Process Design Engineer (Contract)

Sep 2024 – Sep 2025

*MAKEEN Energy EnviroTech, Fredericia & Randers, Denmark*

*(Role extended from internship; see below)*

#### *Fredericia Pyrolysis Plant (Greenfield, FEED Phase)*

- Built steady-state process simulation model in DWSIM; generated stream tables and key outputs to support design reviews
- Developed BFD and progressed to PFD as design matured; reviewed early-stage P&IDs and coordinated comments with system owners and stakeholders
- Prepared equipment datasheets, engineering specifications, and FEED deliverables; managed document version control through SharePoint and ERP systems

#### *Randers Pyrolysis Plant (Operational Facility)*

- Supported PSV revalidation against updated operating scenarios and assumptions in alignment with API 520/521
- Supported nitrogen blanketing calculations for safe shutdown and inert conditions
- Analysed three months of SCADA operational data (millions of rows) using Python to identify downtime patterns and root causes; built reusable automated reporting workflow
- Attended site visits; supported field documentation review and coordination with engineering and

procurement teams

## Process Simulation Engineer (Intern)

Aug 2024 – Sep 2024

*MAKEEN Energy EnviroTech, Fredericia, Denmark*

- Built DWSIM simulation models for the pyrolysis concept and validated results against engineering estimates
- Supported early-stage P&ID review, equipment datasheets, stream tables, and milestone reporting

## Independent Portfolio

Self-initiated work covering areas beyond direct professional experience, documented at [subbu1310.github.io](https://subbu1310.github.io): pump, control valve, and compressor sizing; PSV and flare system sizing (API 520/521/527, Aspen Flare Analyzer); heat exchanger thermal sizing (TEMA); hydraulic calculations; full P&ID development from scratch.

## Education

### MSc Energy Engineering (Thermal and Process Engineering)

2021 – 2024

*Aalborg University, Denmark*

### BTech Mechanical Engineering (Energy Specialisation)

2016 – 2020

*Shiv Nadar University, India*

Class A Scholarship; Dean's List. Published at ICE3 2020 on "Nanofluid and Phase change material applications for PVT modules". UPCST research grant recipient.

## Certifications

- Chemical Process Hazards Analysis with OSHA Safety Standards (HAZOP, FMEA, FTA, What-If, PSM Rule 29 CFR 1910.119)
- PSV and Flare Sizing (API 520/521/527)
- Heat Exchanger Design, Maintenance and Operation
- Pump Sizing and Modelling
- Control Valve Sizing
- AutoCAD P&ID Essential Training
- Compressor Sizing (in progress)

## Languages

English (Fluent) | Tamil and Hindi (Native) | Danish (Actively learning, 2 of 6 modules completed)

## Additional Information

Open to relocation within Denmark and Europe. Willing to travel for project site activities. Eligible to work in Denmark (EU). Volunteer Camp Committee Coordinator, Dansk ICYE (Mar 2025 – present).

## Reference

### David Junkuhn Friis

Process Design Manager, MAKEEN Energy

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